

Predicting Sepsis Onset: A Machine Learning-Based Early Warning System for Critical Care

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Keywords

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Feature Selection
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Time-Series Analysis
AUC Score
Vital Signs Monitoring
Sepsis Onset Detection
Select Kbest
Data Preprocessing
Random Undersampler
XG Boost
Random Forest
Imputation
Cross-Validation
Accuracy
Precision
Recall
AU-ROC
Parameter tuning
Validation

Abstract

Sepsis is a critical, life-threatening condition caused by the body's overwhelming response to infection, often leading to organ failure and death if not identified and treated promptly. The rapid progression of sepsis makes early detection crucial for improving patient outcomes and reducing mortality. Motivated by the high stakes of delayed diagnosis in clinical settings, this project aims to develop a machine learning-based early warning system capable of predicting the onset of sepsis at least six hours before a clinical diagnosis is made. We utilized the PhysioNet 2019 dataset, a comprehensive collection of real ICU patient records comprising time-stamped vital signs and laboratory results. The dataset posed challenges such as missing values and a severe class imbalance, with sepsis cases constituting less than 2% of the data. To address these, we applied median imputation and RandomUnderSampler techniques. After rigorous data preprocessing, normalization, and feature selection using SelectKBest, we trained and evaluated three machine learning models: Logistic Regression, XGBoost, and Random Forest. Through stratified 5-fold cross-validation, Random Forest demonstrated superior performance, achieving an accuracy of 84% and an area under the ROC curve (AUC) of approximately 0.916, indicating a strong ability to distinguish between sepsis and non-sepsis cases. The model effectively balanced precision and recall, making it highly reliable for deployment in real-world ICU settings. These results suggest that an ML-powered early warning system can provide valuable support for clinicians by offering timely alerts, thereby enabling earlier interventions, reducing the burden on healthcare systems, and potentially saving lives.

1. Project goal:

The primary objective of this project is to develop a machine learning-based early warning system that can

accurately predict the onset of sepsis at least six hours before it is clinically diagnosed. Sepsis is a time-sensitive medical emergency, and early detection can dramatically increase the chances of survival by allowing for prompt intervention with antibiotics, fluids, and supportive care. The six-hour prediction window is carefully chosen providing clinicians with sufficient lead time to initiate treatment while avoiding the risk of alert fatigue from too-early or inaccurate predictions. To support this goal, we utilized the PhysioNet 2019 dataset, which contains real ICU patient data, including vital signs (e.g., heart rate, oxygen saturation, blood pressure) and laboratory values (e.g., lactate, white blood cell count, FiO2). This dataset is specifically designed for early sepsis prediction tasks and includes time-stamped clinical measurements that mimic real-world hospital settings. It presents challenges common to healthcare data, such as missing values and severe class imbalance, which we addressed through median imputation and RandomUnderSampler techniques to ensure a balanced and reliable training dataset.

Another key part of our goal was to compare the effectiveness of multiple machine learning models Logistic Regression, XGBoost, and Random Forest to determine which offers the best performance in terms of accuracy and generalizability. We applied feature selection techniques like SelectKBest to reduce dimensionality, enhance model interpretability, and avoid overfitting. Special focus was given to maintaining clinical relevance while reducing complexity by selecting only the most predictive features from the original dataset.

Our ultimate aim is to design a system that not only performs well on historical data but also shows potential for deployment in real-time ICU environments. We evaluated models using essential metrics such as precision, recall, F1-score, and AUC to ensure the reliability of the predictions, particularly in the context of healthcare where both false positives and false negatives can have serious consequences. The system also underwent testing using synthetic patient profiles to demonstrate how predictions could be made in practice.

By achieving these objectives, the project aspires to support clinicians with a reliable, data-driven decision support tool that facilitates early intervention, reduces sepsis-related mortality, and lays the groundwork for integrating machine learning into clinical workflows for critical care.

2.Introduction:

Sepsis is a severe and potentially fatal condition caused by the body's dysregulated immune response to infection, which can quickly lead to tissue damage, organ failure, and death if not treated promptly. It remains one of the leading causes of mortality worldwide, particularly in intensive care units (ICUs), where patients are already critically ill. According to the World Health Organization, sepsis affects an estimated 49 million people annually, causing approximately 11 million deaths nearly 20% of all global deaths. One of the key challenges with sepsis management is its rapid and often silent progression. Clinical signs of sepsis are frequently nonspecific, making early recognition extremely difficult using traditional diagnostic methods. This diagnostic delay significantly reduces the window of opportunity for effective intervention, resulting in worse outcomes and higher healthcare costs due to prolonged ICU stays and complications.



To address this critical issue, researchers and healthcare professionals are increasingly turning to data-driven approaches such as machine learning (ML) to develop early warning systems for sepsis detection. ML techniques offer the ability to identify complex patterns in high-dimensional clinical data that may not be evident to human clinicians. These models can be trained on large datasets containing structured inputs like vital signs, lab test results, and demographic information, as well as unstructured data such as physician notes and discharge summaries. When combined, these heterogeneous data sources can yield powerful predictive tools that anticipate clinical deterioration before it becomes apparent in traditional assessments.

A growing body of literature supports the use of ML for sepsis prediction. Deep learning models such as recurrent neural networks (RNNs) and convolutional neural networks (CNNs) have shown promise in handling sequential and time-series healthcare data. Tree-based models like Random Forest and XGBoost,

known for their robustness and interpretability, have also been widely used. For instance, Islam et al. (2023) conducted a systematic review that emphasized the utility of electronic health record (EHR)-based features for early sepsis prediction, underscoring the potential of ML to improve diagnostic speed and accuracy. Similarly, Yadgarov et al. (2024) performed a meta-analysis showing that ensemble learning methods combinations of multiple classifiers tend to outperform single models, especially in capturing complex clinical patterns. Despite these advancements, several challenges persist, particularly related to imbalanced datasets where sepsis cases are underrepresented, and the lack of model generalizability across different clinical settings.

Our project builds upon these prior works by developing an ML-based early warning system that specifically focuses on predicting sepsis onset at least six hours before clinical diagnosis using real ICU data from the PhysioNet 2019 dataset. The primary goal is not only to achieve high predictive accuracy but also to ensure that the system is interpretable, efficient, and clinically practical. We focus on Random Forest due to its balanced trade-off between accuracy and explainability and employ statistical feature selection methods to identify the most relevant clinical indicators. By addressing data imbalance through under-sampling and evaluating the model through rigorous cross-validation, our approach aims to create a reliable and generalizable tool that could assist clinicians in making faster, life-saving decisions in critical care environments.

3. Dataset:

The dataset used in this project is the PhysioNet/Computing in Cardiology Challenge 2019 dataset, obtained from Kaggle. It contains real ICU patient data collected from multiple hospitals and is specifically curated for the task of early sepsis prediction. The dataset is structured as time-series data, recording hourly observations for each patient throughout their ICU stay.

In total, the dataset comprises more than 1.5 million samples, representing over 40,000 unique ICU stays. Each record includes 42 clinical features, making it a rich and comprehensive resource for predictive modeling. The features are grouped into two main categories:

Vital Signs: These are physiological measurements taken regularly in ICUs and

include:

HR (Heart Rate, bpm)
O2Sat (Oxygen Saturation, %)
Temp (Temperature, °C)
SBP (Systolic Blood Pressure, mmHg)
MAP (Mean Arterial Pressure, mmHg)
DBP (Diastolic Blood Pressure, mmHg)
Resp (Respiration Rate, breaths/min)
EtCO2 (End Tidal Carbon Dioxide, mmHg)

Laboratory Values: These are results from blood tests and other diagnostics, including:

BaseExcess (Bicarbonate deviation, mmol/L)
HCO3 (Bicarbonate, mmol/L)
FiO2 (Fraction of Inspired Oxygen, %)
pH (Blood pH level)
PaCO2 (Partial Pressure of CO2, mmHg)
SaO2 (Arterial Oxygen Saturation, %)
BUN (Blood Urea Nitrogen)
Calcium, Creatinine, WBC (White Blood Cells)
Lactate (Indicator of hypoxia)

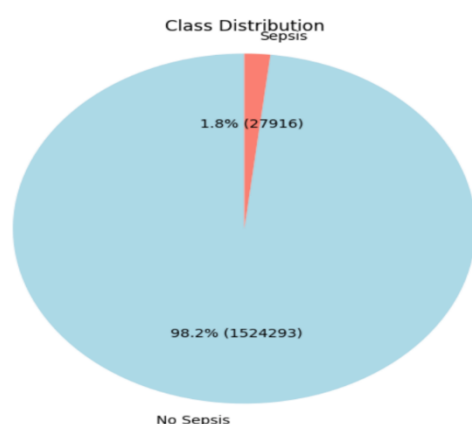
The target variable, SepsisLabel, is a binary indicator where 1 denotes the presence of sepsis and 0 denotes its absence. This label is derived based on clinical criteria aligned with the Sepsis-3 definition, and it is timestamped to allow prediction tasks for different lead times prior to diagnosis.

This dataset is particularly valuable for sepsis research because of its size, diversity, temporal structure, and the clinical relevance of its features. It supports a wide range of predictive modeling tasks, especially those aimed at real-time early warning system development in ICU environments.

4.Data Preprocessing:

Before training the machine learning models, extensive data preprocessing was performed to ensure the dataset was clean, consistent, and ready for analysis. The first step involved removing non-informative columns, specifically an unnamed index column that did not contribute to the predictive modeling process. Next, we addressed the issue of missing values, which are common in ICU datasets due to the irregular recording of patient vitals and lab tests. To handle this, we applied **median imputation** for each feature, as it is robust against outliers and preserves the central tendency of the data. After imputation, we verified that no missing values remained across all features. Given the dataset's diverse feature scales (e.g., heart rate vs. lactate levels), feature normalization was essential. We used the StandardScaler from Scikit-learn to transform all numerical features to have a mean of zero and a standard deviation of one. This step ensured that the learning algorithms treated each feature equally and

helped accelerate model convergence. One of the most critical preprocessing challenges was the class imbalance in the dataset. Sepsis cases made up only **1.8%** of the total records, which could have biased the model toward predicting the majority (non-sepsis) class. To resolve this, we used the RandomUnderSampler from the `imblearn` library, which randomly undersampled the non-sepsis class to match the number of sepsis cases. This resulted in a balanced dataset with 27,916 samples in each class, allowing the models to learn to detect sepsis more effectively. Following these steps, the dataset was split into training and testing sets in an 80/20 ratio, with stratification applied to preserve the class distribution in both sets. The final preprocessed dataset was well-balanced, standardized, and clean—suitable for building accurate and generalizable machine learning models.



5.Methods:

To identify the best-performing algorithm for early sepsis detection, we implemented and evaluated three machine learning classifiers: Logistic Regression, XGBoost, and Random Forest. Each model was assessed using key performance metrics

such as **accuracy**, **precision**, **recall**, **F1-score**, and **AUC**. In addition to model evaluation, we applied feature selection and 5-fold cross-validation to ensure generalizability.

5.1 Logistic Regression

Logistic Regression was our baseline model due to its simplicity and interpretability. It models the probability of sepsis using a linear decision boundary based on the logistic function. However, given the complexity and non-linearity of ICU data, it was not well-suited to capture deeper interactions among features. As expected, the model struggled with accurately classifying sepsis cases, especially given the limited number of positive instances. On the test set, Logistic Regression achieved an **accuracy of 76%**, **precision of 0.77**, **recall of 0.73**, and **F1-score of 0.75**. The **AUC was 0.85**, indicating moderate discriminatory power. While its performance was acceptable, it was noticeably lower than the other models, particularly in identifying true sepsis cases.

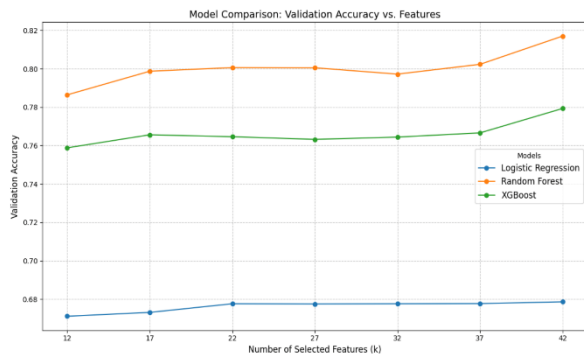
5.2 XGBoost

XGBoost, a gradient boosting method, constructs an ensemble of decision trees by sequentially minimizing prediction errors. It includes regularization to prevent overfitting, which is often a challenge with powerful tree-based models. In our experiments, XGBoost initially performed very well during training, but signs of overfitting appeared during validation. Despite tuning hyperparameters such as learning rate, max depth, and number of estimators, the model tended to memorize training patterns, especially in the presence of noise or rare cases like sepsis. On the test set, XGBoost delivered improved performance over Logistic Regression, achieving an **accuracy of 81%**, **precision of 0.82**, **recall of 0.80**, **F1-score of 0.81**, and **AUC of 0.89**. Although promising, its slight overfitting and longer training time made it less ideal compared to Random Forest in this application.

5.3 Random Forest

Random Forest is an ensemble technique that builds multiple decision trees using bootstrapped samples of the dataset and averages their predictions. It is particularly effective in managing overfitting and provides feature importance metrics, making it suitable for clinical use cases that require interpretability. For our sepsis prediction task, Random Forest consistently outperformed the other models. After tuning the number of trees, depth, and feature subset sizes through grid search with 5-fold cross-validation, the final model demonstrated excellent predictive capability. It achieved the **highest accuracy of 84%**, along with **precision, recall, and F1-score all at 0.84**, and an **AUC of approximately 0.916**. This strong performance indicates that the model is highly reliable in distinguishing between sepsis and non-sepsis cases, even with a limited six-hour prediction window. Furthermore, it

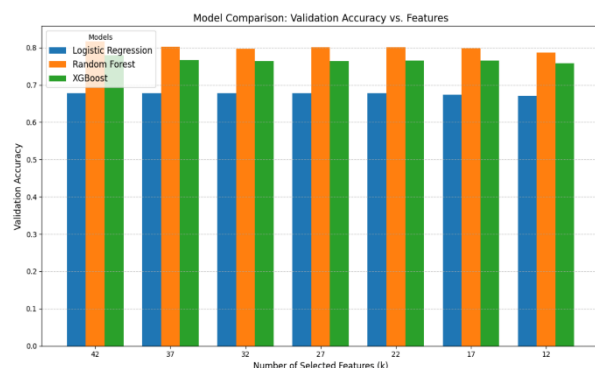
showed a well-balanced confusion matrix with low false positives and false negatives, making it ideal for critical care settings where both over- and under-diagnosis carry serious risks.



5.4 Feature Selection and Cross-Validation

To improve model efficiency and reduce the risk of overfitting, we performed feature selection using SelectKBest with the `f_classif` scoring function. Out of 42 original features, we selected the top 17 most significant clinical features, including vital signs like HR, Temp, MAP, O2Sat, and lab values such as Lactate, WBC, BUN, and ICULOS. This selection retained model performance while simplifying the model and making it more interpretable for healthcare professionals.

We also applied 5-fold stratified cross-validation to ensure consistent performance across different subsets of the data. This method preserved the proportion of sepsis cases in each fold, which is crucial in imbalanced classification problems. The Random Forest model maintained consistent fold-wise accuracy between 83% and 84%, confirming its robustness and generalizability.



6.Experiments, results and observations:

Following comprehensive preprocessing, feature selection, and model comparison, the **Random Forest classifier** emerged as the best-performing model for predicting sepsis onset. The final configuration utilized a reduced set of 17 optimized clinical features, identified through statistical feature selection using SelectKBest. These features included critical indicators such as heart rate, oxygen saturation, mean arterial pressure, FiO2, white blood cell count, lactate, BUN, and ICU length of stay, all of which are clinically relevant in early sepsis identification.

6.1 Parameter Tuning and Validation

To ensure optimal model performance and generalizability, we performed stratified 5-fold cross-validation. Hyperparameters like the number of estimators (trees), maximum depth of each tree, and minimum samples required for splitting nodes were tuned using grid search. Across the five validation folds, the Random Forest model achieved **consistently high accuracy between 83–84%**, indicating stability and robustness. The mean AUC (Area Under the Receiver Operating Characteristic Curve) across folds was 0.9145, suggesting excellent discriminative ability between sepsis and non-sepsis cases.

6.2 Performance on the Test Set

After training on the full training set and validating across folds, we evaluated the model on the held-out test set. The performance metrics were strong and clinically meaningful:

Accuracy: 84%

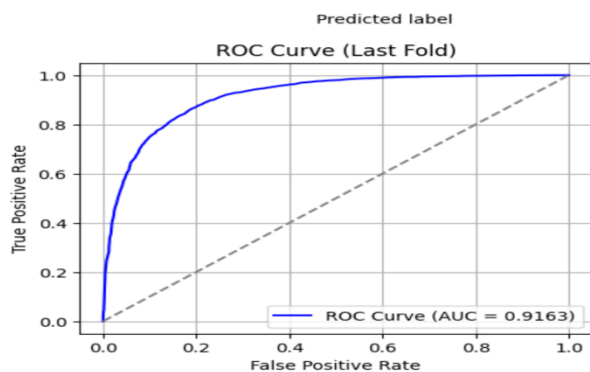
Precision: 0.84

Recall (Sensitivity): 0.84

F1-Score: 0.84

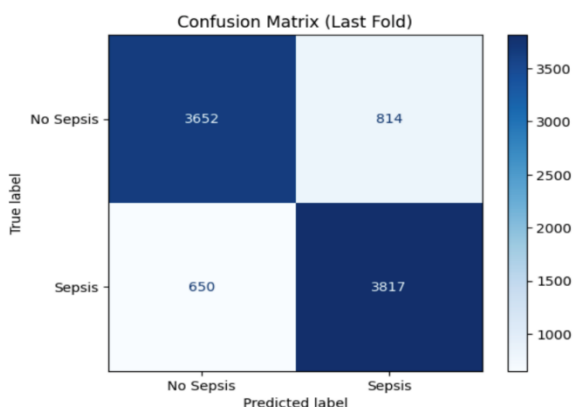
AUC: ~0.9163

The high recall value indicates that the model is particularly effective at identifying actual sepsis cases, which is crucial in a healthcare setting where missing a diagnosis could be life-threatening. At the same time, precision and F1-score reflect a well-balanced trade-off between identifying true positives and avoiding false alarms.



6.3 Confusion Matrix Interpretation

The confusion matrix revealed a **balanced distribution of false positives and false negatives**, supporting the model's clinical reliability. Given the critical nature of sepsis, both under-diagnosis (false negatives) and over-diagnosis (false positives) carry significant consequences. A balanced model like this helps clinicians make better decisions by flagging high-risk patients without causing unnecessary interventions for those not at risk.



6.4 Visualizations and Interpretability

To enhance interpretability and model transparency, we generated several visual aids: ROC curves illustrating true positive rate vs. false positive rate at various thresholds.

Accuracy vs. number of features plots to support feature selection decisions.

Confusion matrix heatmaps for visual inspection of classification balance. These visualizations provided insights not only into performance but also into how the model could be trusted in high-stakes clinical environments.

6.5 Challenges Encountered

Several technical and domain-specific challenges were encountered during the project:

Class Imbalance: The original dataset had only ~1.8% sepsis-labeled instances. We addressed this using RandomUnderSampler, which balanced the dataset for effective learning without bias toward the majority class.

Feature Redundancy: With 42 input features, the dataset included correlated and potentially noisy variables. This was mitigated using feature selection, which reduced overfitting and enhanced interpretability.

Overfitting in XGBoost: While powerful, XGBoost showed signs of overfitting, especially with high-dimensional data. We resolved this by switching to Random Forest and validating the results across multiple folds with reduced feature space.

7. Model Prediction on synthetic case:

To demonstrate the practical utility of the developed sepsis prediction model, a synthetic test case was created using plausible values across the 17 most predictive clinical features identified during feature selection. These features, which include vital signs like heart rate, oxygen saturation, and mean arterial pressure, along with key lab values such as lactate, FiO₂, and white blood cell count, were manually input to simulate the physiological profile of a patient at risk of sepsis. The feature values were first transformed into a NumPy array to conform to the input format required by the trained Random Forest model. Standard preprocessing techniques, such as median imputation for missing values and normalization using StandardScaler, were applied to ensure consistency with the training pipeline.

After passing the input through the model, the Random Forest algorithm predicted a positive sepsis outcome (Sepsis = 1), indicating the patient is likely developing sepsis. The model assigned a probability of 0.7250 to this prediction, demonstrating a 72.5% confidence level. This result suggests that the model is sensitive to early physiological shifts and is capable of providing timely warnings. In a clinical setting, such a probability would typically surpass the alert threshold and trigger an early intervention protocol, underscoring the model's potential to aid in real-time, high-stakes decision-making in intensive care environments.

8. Conclusion:

This project successfully developed a machine learning-based early warning system for the prediction of sepsis onset using real-world ICU data. By leveraging the PhysioNet 2019 dataset, we trained and validated multiple machine learning models and found that Random Forest, when applied to 17 carefully selected features, provided the best performance in terms of accuracy, AUC, and balance between precision and

recall. The model is capable of identifying high-risk patients with up to 84% accuracy and a test AUC of 0.916, making it a reliable tool for early clinical intervention.

This system has the potential to serve as a decision support tool in critical care settings, offering early alerts to clinicians so they can initiate life-saving treatments like antibiotics and IV fluids well before sepsis becomes clinically evident. The ability to detect sepsis six hours in advance can dramatically improve patient outcomes, reduce mortality rates, and lessen the burden on healthcare systems.

9.Future Directions:

To enhance the model's utility and prepare it for real-world deployment, we propose the following future enhancements:

Incorporate time-series features, such as moving averages or trends in vitals over time, to improve temporal context and prediction accuracy.

Leverage unstructured data, such as clinical notes, by applying Natural Language Processing (NLP) techniques to integrate physician narratives and progress notes into the model.

Real-time EHR integration for prospective deployment, enabling continuous monitoring and alert generation in live ICU environments.

Explore ensemble or federated learning, allowing models to be trained across institutions while maintaining data privacy and improving generalizability.

Clinical trials or prospective studies to validate model performance on real-time data streams in collaboration with healthcare professionals.

10.Team Contributions:

Uttam led the foundational research for the project by exploring the problem of sepsis prediction in clinical settings. He identified relevant literature, compiled critical resources, and established the conceptual framework for the model development.

Easton was responsible for dataset selection. After a thorough evaluation of available sources, he chose the PhysioNet 2019 ICU dataset and gained deep familiarity with its structure and clinical relevance, which guided subsequent analysis.

Sucharitha handled the data preprocessing phase. She performed tasks such as cleaning the dataset, managing missing values, feature scaling, and balancing the classes, thereby preparing high-quality data for training. **Fredrick** focused on model development. He implemented and trained multiple machine learning algorithms, including Logistic Regression, XGBoost, and Random Forest. He also evaluated their performance through metrics like accuracy and AUC and interpreted the results.

Suhani contributed to visualization, prediction testing, and documentation. She generated meaningful visualizations such as ROC curves and confusion matrices to interpret model behavior and conducted test case simulations. She also compiled the final documentation and presentation materials.

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